

Package ‘scPipeline’

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Title 'Seurat' and Related R Packages for End-to-End Single Cell Analysis

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Description Reports markers list, differentially expressed genes, associated pathways, cell-type annotations, does batch correction and other related single cell analyses all wrapped within 'Seurat'.

Imports Seurat, batchelor, dplyr, ReactomeGSA, celldex, SingleR, SummarizedExperiment, biomaRt, magrittr, rlang

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AnnotateCellsWithSingleR

Annotate cells in a Seurat object using SingleR with Celldex

Description

This function annotates the cells in a Seurat object using the SingleR package with reference data obtained from the Celldex package.

Usage

```
AnnotateCellsWithSingleR(seurat_object, reference_data = NULL, assay = "RNA")
```

Arguments

`seurat_object` A Seurat object to be annotated.

`reference_data` A reference dataset to use for annotation (e.g., `HumanPrimaryCellAtlasData` from `Celldex`). If `NULL`, `HumanPrimaryCellAtlasData` is used by default.

`assay` The assay in the Seurat object to use for annotation. Default is "RNA".

Value

The Seurat object with cell annotations added to the metadata.

ConvertGeneIdentifiers

Convert Gene Identifiers in a Seurat Object

Description

This function takes a Seurat object with gene identifiers as row names (e.g., RefSeq, Ensembl, Entrez) and converts those identifiers to gene symbols (or Ensembl Gene IDs) using the `biomaRt` package. The function can handle various types of gene identifiers and returns a Seurat object with updated row names.

Usage

```
ConvertGeneIdentifiers(
  seurat_object,
  id_type = "refseq",
  to_id_type = "symbol"
)
```

Arguments

`seurat_object` A Seurat object. The row names of the Seurat object's data or assay slot should represent gene identifiers (e.g., RefSeq, Ensembl, or Entrez IDs).

`id_type` A string specifying the type of the input gene identifiers. Options are: "refseq", "ensembl", "entrez". Default is "refseq".

`to_id_type` A string specifying the type of output gene identifiers. Options are: "symbol", "ensembl". Default is "symbol".

Value

A Seurat object with updated gene names (row names) based on the specified conversion.

Examples

```
## Not run:
# Read 10X counts data from matrix.mtx, barcodes.tsv and genes.tsv
counts <- Read10X(data.dir = "../inst/extdata", gene.column = 1)

# Create Seurat object without batch correction
seurat_obj <- SeuratPreprocess(counts)
seurat_obj <- SeuratLowDim(counts)
# Convert RefSeq IDs to gene symbols
seurat_obj_converted <- ConvertGeneIdentifiers(seurat_obj, id_type = "refseq", \
                                              to_id_type = "symbol")

## End(Not run)
```

ReactomeData

Reactome Data Analysis for Seurat Object

Description

This function performs pathway analysis using ReactomeGSA on a Seurat object with cluster information.

Usage

```
ReactomeData(lowdim_seurat_object)
```

Arguments

```
lowdim_seurat_object
  Seurat object that has clusters information
```

Value

A list containing: - GSVA result (gsva_result) - Pathway expression data (pathway_expression)
- Max difference between pathway expression values (max_difference)

Examples

```
## Not run:
# Read 10X counts data from matrix.mtx, barcodes.tsv and genes.tsv
counts <- Read10X(data.dir = "../inst/extdata", gene.column = 1)

# Create Seurat object without batch correction
seurat_obj <- SeuratPreprocess(counts)
seurat_obj <- SeuratLowDim(counts)
# Reactome Analysis
seurat_reactome <- ReactomeData(seurat_obj)

## End(Not run)
```

SeuratLowDim	<i>Create a Low dimensional Seurat object from scaled seurat object</i>
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Description

This function converts the transformed data to low-dimensional data for downstream analysis.

Usage

```
SeuratLowDim(scaled_seurat_object, ...)
```

Arguments

scaled_seurat_object	A scaled Seurat object.
...	Additional arguments to be passed for downstream analyses.

Value

A Seurat object.

Examples

```
## Not run:
# Read 10X counts data from matrix.mtx, barcodes.tsv and genes.tsv
counts <- Read10X(data.dir = "../inst/extdata", gene.column = 1)

# Create Seurat object without batch correction
seurat_obj <- SeuratPreprocess(counts)
seurat_obj <- SeuratLowDim(counts)

## End(Not run)
```

SeuratMarkers	<i>A thresholded markers list for better calculation of DE genes</i>
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Description

This function calculates differentially expressed genes using Seurat::FindAllMarkers.

Usage

```
SeuratMarkers(lowdim_seurat_object)
```

Arguments

lowdim_seurat_object	Seurat object with cluster information
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Value

A list containing two marker lists: - Full markers list - Thresholded markers list with min.pct = 0.1

Examples

```
## Not run:
# Read 10X counts data from matrix.mtx, barcodes.tsv and genes.tsv
counts <- Read10X(data.dir = "../inst/extdata", gene.column = 1)

# Create Seurat object without batch correction
seurat_obj <- SeuratPreprocess(counts)
seurat_obj <- SeuratLowDim(counts)
# Create Markers list
seurat_markers <- SeuratMarkers(seurat_obj)

## End(Not run)
```

SeuratPreprocess

*Preprocess count data and create a Seurat object***Description**

This function preprocesses count data, optionally applying batch correction using `batchelor::fastMNN`, and creates a Seurat object.

Usage

```
SeuratPreprocess(
  counts_data,
  meta.data = NULL,
  batch_column = NULL,
  use_fastMNN = FALSE,
  ...
)
```

Arguments

<code>counts_data</code>	A matrix or data frame of count data.
<code>meta.data</code>	A data frame containing metadata to include in the Seurat object. Default is <code>NULL</code> .
<code>batch_column</code>	A vector or factor specifying batch assignments for each cell. Default is <code>NULL</code> .
<code>use_fastMNN</code>	Logical. Whether to apply batch correction using <code>fastMNN</code> . Default is <code>FALSE</code> .
<code>...</code>	Additional arguments to be passed to <code>Seurat::CreateSeuratObject</code> .

Value

A Seurat object.

Examples

```
## Not run:  
# Read 10X counts data from matrix.mtx, barcodes.tsv and genes.tsv  
counts <- Read10X(data.dir = "../inst/extdata", gene.column = 1)  
  
# Create Seurat object without batch correction  
seurat_obj <- SeuratPreprocess(counts)  
  
## End(Not run)
```

TransferAnnotations	<i>Transfer annotations to Seurat clusters</i>
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Description

This function assigns cluster-level annotations in a Seurat object based on the majority annotation of cells within each cluster.

Usage

```
TransferAnnotations(seurat_object, annotation_col, cluster_col, output_col)
```

Arguments

seurat_object	Seurat object containing cluster and annotation information.
annotation_col	The name of the metadata column with annotations (character string).
cluster_col	The name of the metadata column with cluster information (character string).
output_col	The name of the output column to store cluster annotations (character string).

Value

The Seurat object with an additional column in its metadata, specified by output_col.

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